

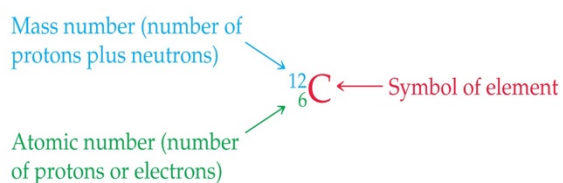
Chapter 21 Nuclear Chemistry

Review
Definitions

Chemical Energy is energy stored in the bonds of atoms and molecules.

Nuclear Energy is due to changes in the nucleus of atoms changing them into different atoms. Nuclear energy is enormous in comparison.

The Nucleus



Nucleus is composed of the two **nucleons**, protons and neutrons. #nucleon = #p⁺ + #n⁰

The number of **protons** is the **atomic number**.

The number of **protons and neutrons** together is the **mass number**.

Isotopes: the same element have the different mass, due to different numbers of neutrons in those atoms.

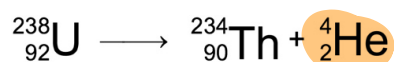
Radioactivity: some nuclides of an elements are unstable, or radioactive, which can be called **radionuclides** (放射性核素).

decay to be more stable

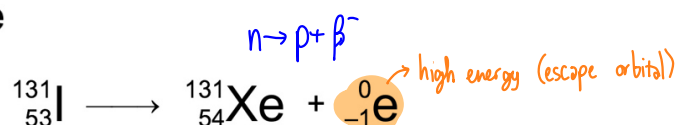
Table 21.1 Properties of Alpha, Beta, and Gamma Radiation			
Property	Type of Radiation		
	α	β	γ
Charge	2+	1-	0
Mass	$6.64 \times 10^{-24}\text{g}$	$9.11 \times 10^{-28}\text{g}$	0
Relative penetrating power	1	100	10,000
Nature of radiation	^4_2He nuclei	Electrons	High-energy photons

Types of Radioactive Decay:

1. **Alpha decay** is the loss of an α -particle.

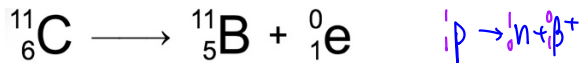


2. **Beta decay** is the loss of a β -particle (a high-speed electron emitted by the nucleus): $^0_{-1}\beta$ or $^0_{-1}\text{e}$



3. **Gamma emission** is the loss of a γ -ray, which is **high-energy radiation** that almost always accompanies the loss of a nuclear particle: $^0_0\gamma$

4. **Positron Emission:** Some nuclei decay by emitting a **positron**, a particle that has the same mass as, but an opposite charge to, that of an electron:



5. **Electron Capture (k-Capture):** An electron from the surrounding electron cloud is absorbed into the nucleus.

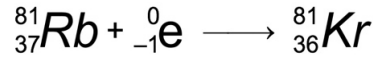


Table 21.3 Types of Radioactive Decay

Type	Nuclear Equation	Change in Atomic Number	Change in Mass Number
Alpha decay	${}^A_Z\text{X} \longrightarrow {}^{A-4}_{Z-2}\text{Y} + {}^4_2\text{He}$	-2	-4
Beta emission	${}^A_Z\text{X} \longrightarrow {}^A_{Z+1}\text{Y} + {}^0_{-1}\text{e}$	+1	Unchanged
Positron emission	${}^A_Z\text{X} \longrightarrow {}^A_{Z-1}\text{Y} + {}^0_{+1}\text{e}$	-1	Unchanged
Electron capture*	${}^A_Z\text{X} + {}^0_{-1}\text{e} \longrightarrow {}^A_{Z-1}\text{Y}$	-1	Unchanged

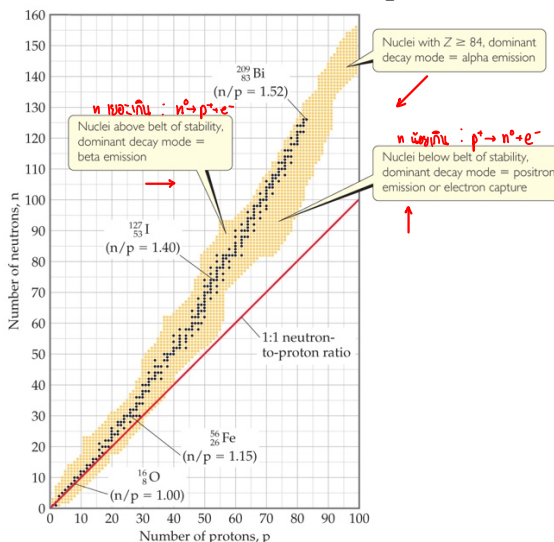
*The electron captured comes from the electron cloud surrounding the nucleus.

Table 21.2 Particles Found in Nuclear Reactions

Particle	Symbol
Neutron	${}^1_0\text{n}$ or n
Proton	${}^1_1\text{H}$ or p
Electron	${}^0_{-1}\text{e}$
Alpha particle	${}^4_2\text{He}$ or α
Beta particle	${}^0_{-1}\text{e}$ or β^-
Positron	${}^0_{+1}\text{e}$ or β^+

Nuclear stability:

- Repulsions between the protons in the nucleus. *needed more n_0 to counteract*
- Strong nuclear force** helps keep the nucleus together.
- The ratio of neutrons to protons. *$\rightarrow e^-$ capture / positron emission*



Stable or unstable?

- belt of stability**--it shows what nuclides would be stable.
- Unstable: **Nuclei above this belt have too many neutrons**--decay by emitting **beta particles**. Nuclei below the belt have too many protons--become more stable by positron emission or electron capture.

#p + 1 = n → isotope 140

1 H (2)																	2 He (2)	
3 Li (2)	4 Be (1)																	10 Ne (3)
11 Na (1)	12 Mg (3)																	18 Ar (3)
19 K (2)	20 Ca (5)	21 Sc (1)	22 Ti (5)	23 V (2)	24 Cr (4)	25 Mn (1)	26 Fe (4)	27 Co (1)	28 Ni (5)	29 Cu (2)	30 Zn (5)	31 Ga (2)	32 Ge (4)	33 As (1)	34 Se (5)	35 Br (2)	36 Kr (6)	
37 Rb (1)	38 Sr (3)	39 Y (1)	40 Zr (4)	41 Nb (1)	42 Mo (6)	43 Tc (0)	44 Ru (7)	45 Rh (1)	46 Pd (6)	47 Ag (2)	48 Cd (6)	49 In (1)	50 Sn (10)	51 Sb (2)	52 Te (6)	53 I (1)	54 Xe (9)	

Number of stable isotopes

Elements with two or fewer stable isotopes

Elements with three or more stable isotopes

3. **Magic numbers** of 2, 8, 20, 28, 50, or 82 protons or 2, 8, 20, 28, 50, 82, or 126 neutrons result in more stable nuclides.

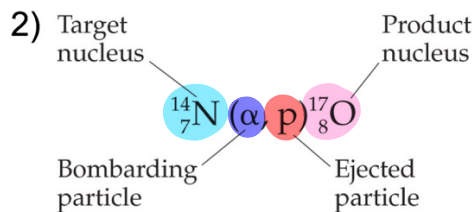
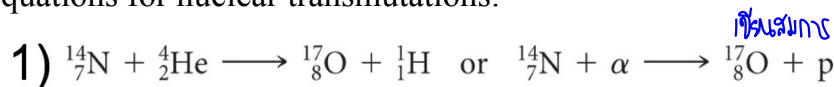
4. Nuclei with an even number of protons and neutrons tend to be more stable than those with odd numbers.

Note: There are no stable nuclei with an atomic number greater than 83. Nuclei with such large atomic numbers tend to decay by alpha emission.

Nuclear transmutations can be induced by accelerating a particle to collide it with the nuclide. ^{n° / other nucleus} ^{forming radioisotopes} ^{overcome electrostatic repulsion}

Particle accelerators ("atom smashers") are enormous, having circular tracks with radii that are miles long. ^{to accelerate particles and create charged particles} ^{no not repelled → no need for acceleration}

Nuclear equations for nuclear transmutations:



Radioactive decay is a first-order process.

Half-life is the time required for half of a radionuclide sample to decay.

Rate = kN $\frac{0.693}{k} = t_{1/2}$ ^{unaffected by T, P, state}

Measuring Radioactivity ^{→ # disintegration per unit time}

Activity is the rate at which a sample decays.

Unit: ^{SI Unit} Becquerel (Bq): one disintegration per second.

Curie (Ci): 3.7×10^{10} disintegrations per second, which is the rate of decay of 1 g of radium.

Instruments:

1. **Film badges** are used by people who work with radioactivity to measure their own exposure over time. ^{Fogging of photographing plates (darker)}
2. **Geiger counter** measures the amount of activity present in a radioactive sample. ^{Ionizing matter → conduction of electrical current}
3. Phosphors (scintillation counters): phosphors absorb radioactivity and emit light. **Scintillation counter** used to measure the amount of light emitted by a phosphor. It converts the light to an electronic response for measurement.

Radiotracers are ^{follow} radioisotopes used to study a chemical reaction. ^{trace the path}

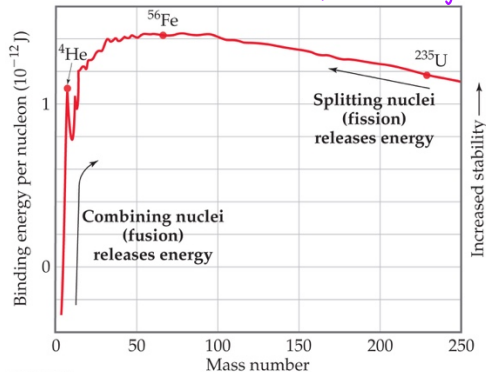
Energy in nuclear reaction: $E = mc^2$.

$\Delta E = c^2 \Delta m$
 $\rightarrow m_{\text{products}} - m_{\text{reactants}}$
 ^(มวลรวมลดลง)

^{C-14} ^{คาร์บอน-14} ^{half life ≈ 5700 yr}
 ^(Ratio C-14 : C-12 ในธรรมชาติ)
 ^{มวล → มวล C-14 → decay ตามอัตราคงที่ อยู่ประมาณเท่าไร}
 ^(ใช้วัดอายุโบราณวัตถุ) ^{= 100 ปี}

The masses of nuclei are always less than those of the individual parts. This mass difference is called the **mass defect**.

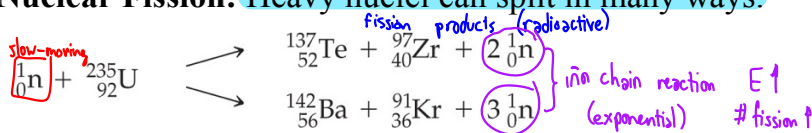
The energy needed to separate a nucleus into its nucleons is called the **nuclear binding energy**.



more Heavy nuclei gain stability and give off energy when they split into two smaller nuclei. This is **fission**.

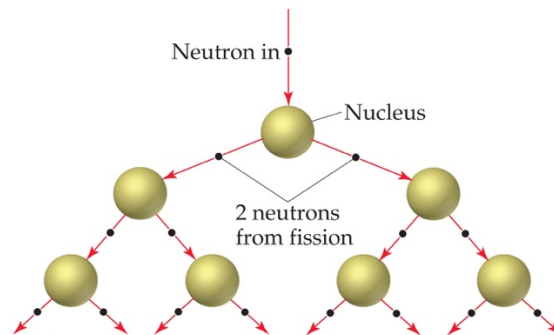
being more stable Lighter nuclei emit great amounts of energy by being combined in **fusion**.

Nuclear Fission: Heavy nuclei can split in many ways.

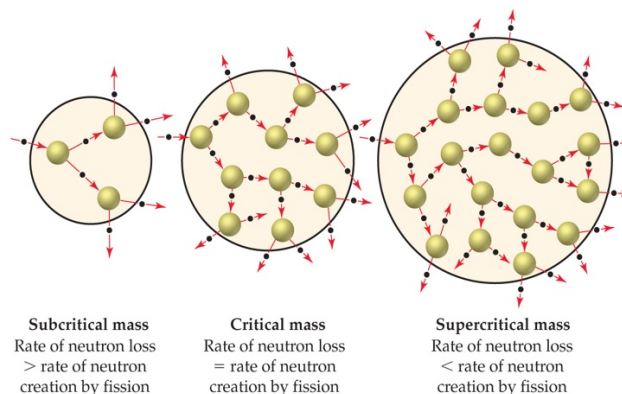


Nuclear chain reaction is a process continues that

1. Bombardment of the radioactive nuclide with a neutron starts the process.
2. Neutrons released in the transmutation strike other nuclei, causing their decay and the production of more neutrons.

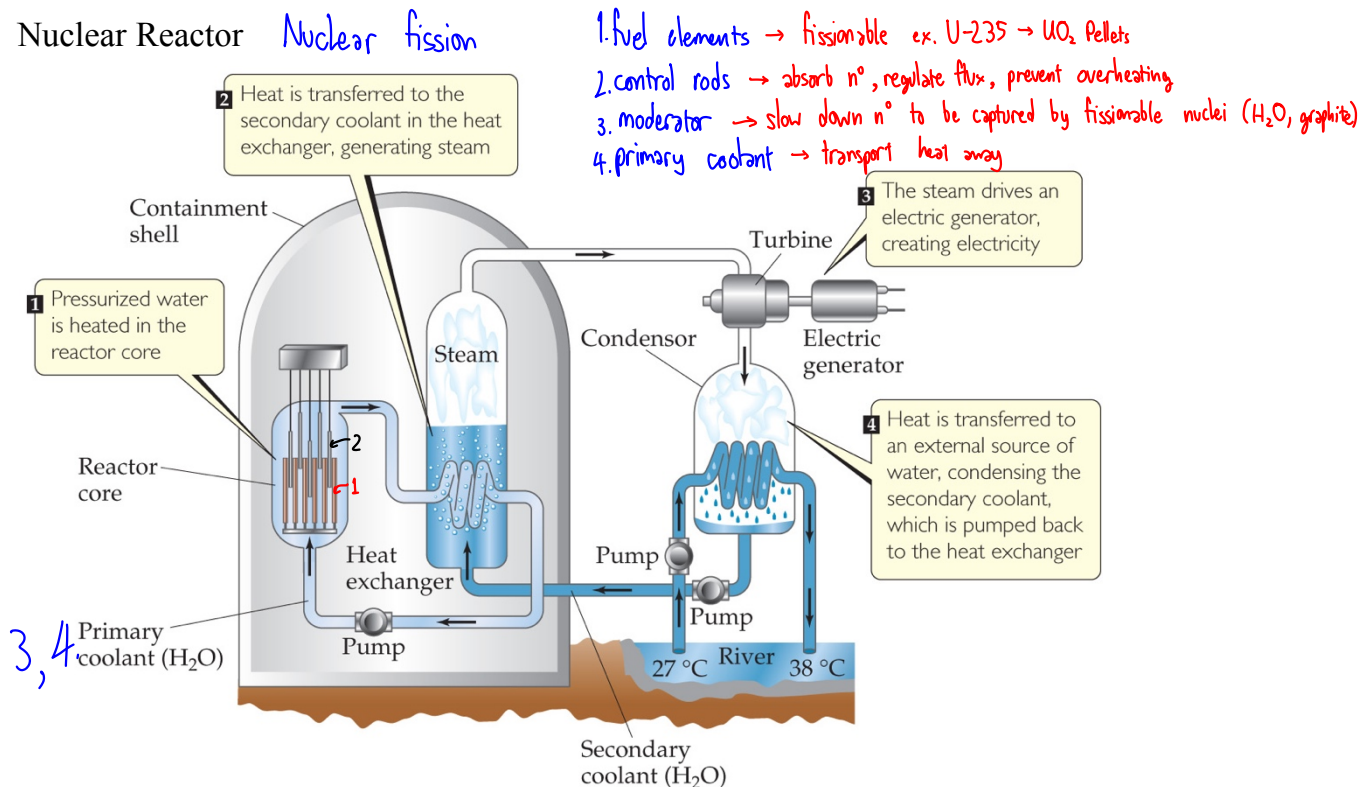


The minimum mass that must be present for a chain reaction to be sustained is called the **critical mass**. If more than critical mass is present (**supercritical mass**), an explosion will occur.



Nuclear Reactor

Nuclear fission



Reactors must be stopped periodically to replace or reprocess the nuclear fuel. The original intent was that **nuclear waste** would then be transported to reprocessing or storage sites.

$\hookrightarrow$ decrease eff. of reactor by capturing n°

Nuclear Fusion—small atoms are combined, much energy is released.

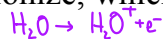
Extremely high temperatures and pressures are needed to cause nuclei to fuse, which was achieved using an **atomic bomb to initiate fusion in a hydrogen bomb**.



make atoms excited/ionized, water absorbs most of radiation

Ionizing radiation is more harmful to living systems than nonionizing radiation (such as radiofrequency electromagnetic radiation).

Ionizing radiation---Most living tissue is ~70% water, ionizing radiation is that which causes water to ionize, which creates unstable, very reactive OH radicals, then damage cells.



$\hookrightarrow$ free radical (attacking biomolecules)

Low-level, long-term **exposure** can cause health issues. Damage to the growth-regulation mechanism of cells results in cancer.

Radiation dose:

SI Gray (Gy): absorption of 1 J of energy per kg of tissue.

Rad (for radiation absorbed dose): absorption of 0.01 J of energy per kg of tissue (100 rad = 1 Gy).

Note: Not all forms of radiation harm tissue equally. A **relative biological**

PET Scans: use isotopes that emit positron to monitor blood flow, O_2 & metabolic rates, brain
 $\hookrightarrow$ positron emission tomography

α : stopped by skin
 β : 1 cm beyond skin
 γ : through tissue

Gamma: 1
 Alpha: 10
 SI: Sievert (1 Sv = 100 rem)

effectiveness (RBE) is used to show how much biological effect there is.

The **effective dose** is called the **rem** for medicine $= \text{rad} \times \text{RBE}$

Table 21.9 Effects of Short-Term Exposures to Radiation

Dose (rem)	Effect
0–25	No detectable clinical effects
25–50	Slight, temporary decrease in white blood cell counts
100–200	Nausea; marked decrease in white blood cell counts
500	Death of half the exposed population within 30 days

Radon-222 is a decay product of uranium-238, which is found in rock formations and soil. Most of the decay products of uranium remain in the soil, but radon is a gas.